

Section 2.3

Exercise (2.3.1). Find all subgroups of $Z_{45} = \langle x \rangle$, giving a generator for each. Describe the containments between these subgroups.

Solution: Recall that by properties of finite cyclic groups since $|\langle x \rangle| = n$ then for each integer divisor a dividing n there exists a unique subgroup of $\langle x \rangle$ order a generated by $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore we know that every subgroup of $\langle x \rangle$ corresponds bijectively to the positive divisors of n .

Enumerating the divisors of 45,

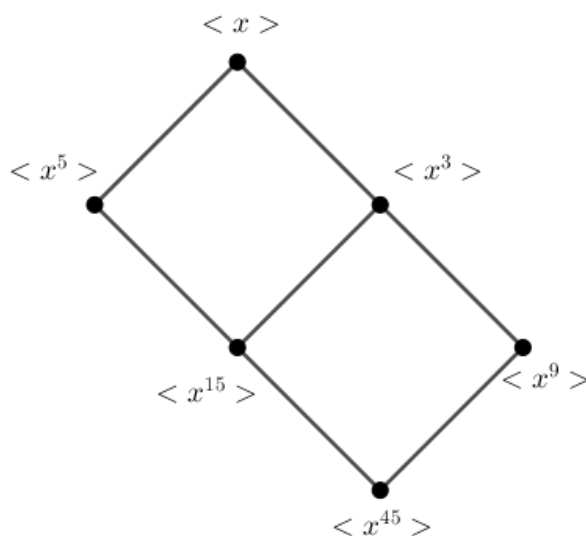
$$1, 3, 5, 9, 15, 45$$

Each corresponding bijectively to the group generated by $\langle x^d \rangle$, where $d = \frac{n}{a}$

$$\langle x^{45} \rangle, \langle x^{15} \rangle, \langle x^9 \rangle, \langle x^5 \rangle, \langle x^3 \rangle, \langle x \rangle$$

Note that since $3 \mid 9, 15$ and $5 \mid 15$ we get the following lattice diagram,

FIGURE 1. Lattice Diagram for $Z_{45} = \langle x \rangle$



□

Exercise (2.3.2). If x is an element of the finite group G and $|x| = |G|$, prove that $G = \langle x \rangle$. Give an explicit example to show that this result need not be true if G is an infinite group.

Proof: Suppose x is an element of the finite group G such that $|x| = |G| = n$. Since $|x| = n$ we know that, $1, x, x^2, \dots, x^{n-1}$ are all distinct, since if there were two powers $x^j = x^i$ with $i < j < n$ then $x^{j-i} = 1$ and then $|x| \neq n$. Note that the distinct powers $1, x, x^2, \dots, x^{n-1}$, comprise $\langle x \rangle$ we know that $\langle x \rangle$ contains n distinct elements, all powers of x which must also be contained in G . Thus $\langle x \rangle = G$.

Consider the arithmetic group under $\mathbb{Z}$, and let $x = 2$. The order of both x and the set is infinite, however $\langle 2 \rangle$ generates only positive integers. $\square$

Exercise (2.3.3). Find all generators for $\mathbb{Z}/48\mathbb{Z}$.

Solutions: By Proposition 6 we know that for some finite group H with generator $\langle x \rangle$ whose order $|x| = n$ then $H = \langle x^a \rangle$ if $(a, n) = 1$. Clearly $\mathbb{Z}/48\mathbb{Z}$ has generator $\langle \bar{1} \rangle$ whose order is by definition $|\langle \bar{1} \rangle| = 48$. Therefore all residue classes that generate $\mathbb{Z}/48\mathbb{Z}$ are relatively prime to 48. Therefore $\bar{1}, \bar{5}, \bar{7}, \bar{11}, \bar{13}, \bar{17}, \bar{19}, \bar{23}, \bar{25}, \bar{29}, \bar{31}, \bar{35}, \bar{37}, \bar{41}, \bar{43}, \bar{47}$ Generate $\mathbb{Z}/48\mathbb{Z}$. $\square$

Exercise (2.3.5). Find the number of generators for $\mathbb{Z}/49000\mathbb{Z}$.

Solution: As we recalled in the previous problem, every residual class whose representative is coprime to the order of the group generates the the group. Using Euler's φ – *function* we can find the number of elements from $[49000]$ which are coprime to 49000. Determining the prime factorization of 49000,

$$49000 = 49(10^3) = (7^2)(5^3)(2^3)$$

Applying Euler's φ – *function* we get,

$$\varphi(49000) = 7(6)5^2(4)2^2(1) = 16800.$$

$\square$

Exercise (2.3.8). Let $Z_{48} = \langle x \rangle$. For which integers a does the map ϕ_a defined by $\phi_a : \bar{1} \rightarrow x^a$ extend to an isomorphism from $\mathbb{Z}/48\mathbb{Z}$ into Z_{48} .

Proof:

□

Exercise (2.3.9). Let $Z_{36} = \langle x \rangle$ For which integers a does the map ϕ_a defined by $\phi_a : \bar{1} \rightarrow x^a$ extend to a well defined homomorphism from $\mathbb{Z}/48\mathbb{Z}$ into Z_{48} .

Proof:

□

Exercise (2.3.16). Assume $|x| = n$ and $|y| = m$. Suppose that x and y commute. Prove that $|xy|$ divides the least common multiple of m and n . Need this be true if x and y do not commute? Give an example of commuting elements x, y such that the order of xy is not equal to the least common multiple of $|x|$ and $|y|$?

Proof: Let $|x| = n$ and $|y| = m$ and suppose that x and y commute. Let $i = \text{lcm}(m, n)$, therefore there exists some $j, k \in \mathbb{Z}$ such that $mj = i$ and $nk = i$. Since x and y commute,

$$(xy)^i = x^i y^i = x^{nk} y^{mj} = 1^k 1^j = 1.$$

Since $(xy)^i = 1$ by Proposition 3 it follows that $|xy| \mid i$. For an example where commuting elements x, y such that the order of xy is not equal to the least common multiple of $|x|$ and $|y|$ consider the $\mathbb{Z}_4/\mathbb{Z}$ and note that $|\bar{1}| = 4$ and $|\bar{3}| = 4$ and therefore the least common multiple is 4 but $|\bar{3}\bar{1}| = |\bar{4}| = 1 \neq 4$. □

Exercise (2.3.23). Show that $(\mathbb{Z}/2^n\mathbb{Z})$ is not cyclic for any $n \geq 3$ [Find two distinct subgroups of order 2].

Proof: To show that $(\mathbb{Z}/2^n\mathbb{Z})$ is not cyclic it is sufficient to show that $(\mathbb{Z}/2^n\mathbb{Z})$ contains two distinct subgroups of order two. Meaning there are two subgroups, H, G of $(\mathbb{Z}/2^n\mathbb{Z})$ with elements $H = 1, h$ and $G = 1, g$ such that $h \neq g$. This property demonstrates that there is no single $x \in (\mathbb{Z}/2^n\mathbb{Z})$ for which $h, g \in \langle x \rangle$ and therefore $(\mathbb{Z}/2^n\mathbb{Z}) \neq \langle x \rangle$. Let $n \geq 3$, and consider $G = 1, 2^n + 1$ and $H = 1, 2^n - 1$. Note

that,

$$gg = (2^{n-1}+1)^2 = (2^{n-1}(2^{n-1})+(2)2^n+1^2) = (2^{2n-2}+(2)2^n+1^2) = 2^n(2^{n-2}(-2)+2)+1 \cong 1 \pmod{2^n}$$

$$hh = (2^{n-1}-1)^2 = (2^{n-1}(2^{n-1})-(2)2^n+1^2) = (2^{2n-2}-(2)2^n+1^2) = 2^n(2^{n-2}(-2)-2)+1 \cong 1 \pmod{2^n}$$

Note that $2^{n-1} + 1 \neq 2^{n-1} - 1$ since $n \geq 3$. Therefore H and G are two distinct subgroups of order 2. Thus $(\mathbb{Z}/2^n\mathbb{Z})$ is not cyclic. $\square$

Section 2.4

Exercise (2.4.1). Prove that if H is a subgroup of G then $\langle H \rangle = H$.

Proof: Suppose H is a subgroup of G , and consider the subgroup of G generated by the set H , $\langle H \rangle$. By definition

$$\langle H \rangle = \bigcap_{H \subseteq H \text{ and } H \leq G} H.$$

$\langle H \rangle$ is the intersection of all subgroups of G containing the set H . However H is a subgroup, and therefore any subgroup of G which contains an element from H must contain the whole set H . Thus intersection of all subgroups of G containing the set H is the group H . $\square$

Exercise (2.4.2). Prove that if A is a subset of B then $\langle A \rangle \leq \langle B \rangle$. Give an example where $A \subset B$ with $A \neq B$ but $\langle A \rangle = \langle B \rangle$.

Proof: Let G be a group such that $A, B \subseteq G$. Suppose $A \subseteq B$ and consider $a \in \langle A \rangle$.

Since $A \subseteq B$ we know that $a \in B$ and clearly if $a \in B$ it will be in the intersection of all the subgroups of G which contain B . Thus $a \in \langle B \rangle$.

For an example consider, the group $G = \mathbb{Z}_4/\mathbb{Z}$ and let $A = \{\bar{1}\}$ and $B = \{\bar{1}, \bar{2}\}$ clearly $A \neq B$ but since G is cyclic, any subgroup which contains $\bar{1}$ will generate the whole group, hence $\langle A \rangle = \langle B \rangle$. $\square$

Exercise (2.4.4). Prove that if H is a subgroup of G then H is generated by the set $H - 1$.

Proof: Suppose H is a subgroup of G and consider $\langle H - 1 \rangle$. By definition $\langle H - 1 \rangle$ is the intersection of every subgroup in G which contains the set $H - 1$. Consider that any such subgroup S which contains the set $H - 1$ contains some elements $h, h^{-1} \in H$ such that $h, h^{-1} \neq 1$ since H is a subgroup. Since S is a subgroup it must also contain $hh^{-1} = 1$. Therefore every subgroup in G which contains the set $H - 1$ must also contain 1 and thus the intersection is H . $\square$

Section 2.5

Exercise (2.5.5). Use the given lattice to find all elements $x \in D_{16}$ such that $D_{16} = \langle x, s \rangle$

Proof: Recall that we can enumerate the elements of D_{16} like so,

$$D_{16} = \{r, s | r^8 = s^2 = 1, rs = sr^{-1}\}.$$

Note that $\langle s, r^2 \rangle$ by the lattice diagram is a subgroup of D_{16} . Since $2|6$ we know that $\langle s, r^2 \rangle = \langle s, r^6 \rangle$ and by the lattice diagram $\langle s, r^4 \rangle$ is a subgroup of $\langle s, r^2 \rangle$. Also note that $\langle s, sr^2 \rangle$, $\langle s, sr^4 \rangle$, and $\langle s, sr^6 \rangle$ are contained within $\langle s, r^2 \rangle$.

By definition we know that $x = r$ gives us $\langle r, s \rangle$ generates the group D_{16} . Because

$$(r^3)^3 = (r^5)^5 = (r^7)^7 = r,$$

we know that

$$\langle r \rangle = \langle r^3 \rangle = \langle r^5 \rangle = \langle r^7 \rangle,$$

so $\langle s, r^3 \rangle$, $\langle s, r^5 \rangle$, and $\langle s, r^7 \rangle$ generate the group D_{16} . Similarly we know that $\langle s, sr \rangle$ contains $s(sr) = r$ and hence $\langle s, sr \rangle$ generates D_{16} . By the same argument it follows that $\langle s, sr^3 \rangle$, $\langle s, sr^5 \rangle$, and $\langle s, sr^7 \rangle$ generate D_{16} . Therefore $x \in \{r, r^3, r^5, r^7, sr, sr^3, sr^5, sr^7\}$ are all the elements $x \in D_{16}$ such that $D_{16} = \langle x, s \rangle$ $\square$

Exercise (2.5.6a). Use the given lattices to help find the centralizer of every element in D_8

Solution: □

Exercise (2.5.6c). Use the given lattices to help find the centralizer of every element in S_3

Solution: □

Exercise (2.5.8). *Solution:* □

Exercise (2.5.9). *Solution:* □

Section 3.1

Exercise (3.1.1). Let $\varphi : G \rightarrow H$ be a homomorphism and let E be a subgroup of H . Prove that $\varphi^{-1}(E) \leq G$. If $E \triangleleft H$ prove that $\varphi^{-1}(E) \triangleleft G$. Deduce that $\ker \varphi \triangleleft G$.

Proof: Suppose $\varphi : G \rightarrow H$ is a homomorphism and consider E , a subgroup of H . By properties of homomorphisms $1 \in \varphi^{-1}(E)$. Now consider $x, y \in \varphi^{-1}(E)$. By definition there exists some $a, b \in E$ such that $\varphi(x) = a$, and $\varphi(y) = b$. Consider xy and note that since φ is a homomorphism, $\varphi(xy) = \varphi(x)\varphi(y) = ab$. Since E is a group $ab \in E$ and therefore by definition $xy \in \varphi^{-1}(E)$. By properties of homomorphisms we know that $\varphi(x^{-1}) = \varphi(x)^{-1} = a^{-1}$, since $a^{-1} \in E$ we know that $x^{-1} \in \varphi^{-1}(E)$. Thus $\varphi^{-1}(E) \leq G$.

Suppose $E \triangleleft H$. By definition $heh^{-1} \in E$, for all $h \in H$ and $e \in E$. Let $g \in G$ and $a \in \varphi^{-1}(E)$. Note $\varphi(gag^{-1}) = \varphi(g)\varphi(a)\varphi(g)^{-1} = heh^{-1}$. Since $E \triangleleft H$, $heh^{-1} \in E$ and therefore $\varphi^{-1}(E) \triangleleft G$.

Let E be the trivial subgroup which is also normal in H . It follows that $\varphi^{-1}(E) = \ker \varphi$, and by the previous result $\ker \varphi \triangleleft G$. □

Exercise (3.1.3). Let A be an abelian group and let B be a subgroup of A . Prove that A/B is abelian. Give an example of a non-abelian group G containing a proper normal subgroup N such that G/N is abelian.

Proof: Suppose A be an abelian group and let B be a subgroup of A . Consider $x, y \in A$, and note that $xB, yB \in A/B$. Since B is a subgroup of A we know that $xB yB = (xy)B$, and since A is abelian,

$$xB yB = (xy)B = (yx)B = yB xB.$$

Therefore A/B is abelian. □

Exercise (3.1.4). Prove that in the quotient group G/N , $(gN)^\alpha = g^\alpha N$ for all $\alpha \in \mathbb{Z}$.

Proof: Suppose N is a subgroup of G and consider the quotient group G/N . By Proposition 5 we know that, $g_1N \cdot g_2N = (g_1g_2)N$ is a well defined operation. what follows is that $(gN)^{-1} = g^{-1}N$. Note what follows,

$$(gN)^\alpha = g_1N \cdot g_2N \cdot \dots \cdot g_\alpha N = (g_1g_2\dots g_\alpha)N = g^\alpha N.$$

□

Exercise (3.1.5). Use the preceding exercise to prove that the order of the element gN in G/N is n , where n is the smallest positive integer such that $g^n \in N$.

Proof: Suppose $gN \in G/N$, and let n be the smallest positive integer such that $g^n \in N$. By the previous problem we know that $(gN)^n = g^n N = N$ since $g^n \in N$. Thus $|gN| \leq n$. Suppose that $(gN)^i = N$, by definition it follows that $(gN)^i = g^i N = N$, therefore $g^i \in N$. Note that n is the smallest such integer such that and therefore $|gN| \geq n$. Thus $|gN| = n$. □

Exercise (3.1.7). Define $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x + y$. Prove that π is a surjective homomorphism and describe the kernel and fibers of π geometrically

Proof: Suppose $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x + y$. Let $(x, y), (a, b) \in \mathbb{R}^2$ and consider $\pi((x, y) + (a, b)) = \pi((a + x, b + y)) = a + x + b + y = a + b + x + y = \pi((a, b))\pi((x, y))$. Thus π is a homomorphism. Let $x \in \mathbb{R}$, consider $(x, 0) \in \mathbb{R}^2$ and note $\pi((x, 0)) = x + 0 = x$ and thus π is a surjection.

Now consider the kernel of π , and note that $\ker(\pi) = \{(x, -x) \in \mathbb{R}^2 | x \in \mathbb{R}\}$ since $\pi(x, -x) = x - x = 0$ which is the identity in $\mathbb{R}$. Geometrically this is the line $y = -x$. Recall that the fibers of π for a given element $a \in \mathbb{R}$ is the set of all $\phi^{-1}(a) = \{(x, y) \in \mathbb{R}^2 : x + y = a\}$. Thus fibers of π are of the form $y = a - x$, and geometrically represent lines parallel to the kernel. $\square$

Exercise (3.1.22). a. Prove that if H and K are normal subgroups of group G then their intersection $H \cap K$ is also a normal subgroup of G .

Proof: Suppose H and K are normal subgroups of group G . Suppose $a, b \in H \cap K$ clearly $ab^{-1} \in H \cap K$ since H and K are subgroups and both contain a, b . Hence $H \cap K$ is a subgroup. Let $g \in G$ and $a \in H \cap K$ clearly $gag^{-1} \in H$ since H is normal in G , and $gag^{-1} \in K$ since K is normal in G . Thus $gag^{-1} \in H \cap K$ and therefore $H \cap K$ is a normal subgroup of G . $\square$

b. Prove that the intersection of an arbitrary nonempty collection of normal subgroups of a group is a normal subgroup.

Proof: Suppose $\mathcal{H}$ is a family of nonempty normal subgroups in a group G . Consider the following $a \in \cap \mathcal{H}$. Clearly for $g \in G$, $gag^{-1} \in H_i$ for all $H_i \in \mathcal{H}$ since $\mathcal{H}$ is a family of normal subgroups. Thus by definition it follows that, $gag^{-1} \in \cap \mathcal{H}$. $\square$

Exercise (3.1.24). Prove that if N is a normal subgroup of G and H is any subgroup of G then $N \cap H$ is a normal subgroup of H .

Proof: Suppose N is a normal subgroup of G and H is any subgroup of G . Let $a, b \in N \cap H$, clearly $ab^{-1} \in N \cap H$, hence $N \cap H \leq H$. Note that since N is normal in G and H

is a subgroup for all $h \in H$ $hah^{-1} \in N$. Since $a \in H$ and H is a subgroup we know that $hah^{-1} \in H$. Thus $hah^{-1} \in N \cap H$ and therefore $N \cap H$ is normal in H . $\square$

Exercise (3.1.28). Let N be a finite subgroup of a group G and assume $N = \langle S \rangle$ for some subset S of G . Prove that an element $g \in G$ normalizes N if and only if $gSg^{-1} \subseteq N$.

Proof: Let N be a finite subgroup of a group G and $N = \langle S \rangle$ for some subset S of G .

Suppose $g \in G$ normalizes N . By definition it follows that for $g \in G$, $gng^{-1} \in N$. By definition N is also the intersection of all subgroups containing S , hence $S \subseteq N$ and since S is a subset of G for all $s \in S$, $gsg^{-1} \in N$. Thus $gSg^{-1} \subseteq N$.

Now suppose $gSg^{-1} \subseteq N$. Since N is the group generated by S by definition every n is equal to some unique, finite product of the elements of S , hence

$$n = s$$

$\square$